

# Isaac Mubiru

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## PROFESSIONAL SUMMARY

Senior backend engineer (Node.js + TypeScript, 5+ years) who owns systems end-to-end — from scoping and architecture to high-reliability delivery. Built distributed Kafka + NATS + Kubernetes microservices for 4 NGO tenants at CodeBits and led a zero-downtime migration of 20+ production apps from MongoDB to Firestore at MTailor. Adjacent AI exposure: integrated OpenAI + Google Vision into a WhatsApp healthcare chatbot for prescription OCR. Comfortable shipping scalable APIs and developer tooling across Product, Frontend and Design.

## CORE COMPETENCIES

TypeScript + Node.js

Distributed systems

Scalable APIs

Microservices (Kafka/NATS/K8s)

End-to-end ownership

Backend infra + tooling

Cross-functional delivery

LLM integration (adjacent)

## WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer (US Remote) — reporting to CTO

- Owned a **zero-downtime migration of 20+ production Node.js apps** end-to-end — scoping, architecture, parallel cutover, observability
- Built **real-time two-way sync between MongoDB and Firestore using Node.js + Google Pub/Sub** with idempotent consumers and drift detection
- Implemented a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS via a Python script with retries
- Implemented Express Shipping integration, generating \$40 additional revenue per order
- Saved the company \$5,000/month by migrating all services off AWS

CodeBits

January 2020 – July 2021

Software Engineer / Lead (Uganda Remote)

- Led a team of 4 developers building legal-tech systems for NGOs in Uganda
- Architected a microservices backend with Apache Kafka, Docker, Kubernetes, NATS Streaming, and gRPC** for FIDA Uganda — 4 NGO tenants on shared infrastructure with typed contracts
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda
- Developed a cross-platform mobile app for LASPNET using React Native and Expo
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC

Mind2matter

January 2022 – July 2022

Software Engineer (US Remote) — reporting to CTO

- Built backends for DeFi applications using Web3 and Node.js — wallet signing, contract calls
- Delivered React UIs from Figma designs under tight timelines with daily cross-functional syncs

Dr Wealth

August 2021 – January 2022

Software Engineer (Singapore Remote)

- Optimized a PWA delivering real-time stock prices from a Firebase backend
- Built ad-hoc jobs to query over 2 million Firestore records keeping customer prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js hosted on Heroku

MTailor

March 2022 – July 2022

Contractor (US Remote)

- Built a Node.js migration tool to read MongoDB and write to Firestore
- Wrote SDK documentation for fellow engineers and trained the Ops team on Firebase

## PROJECTS

**Healthcare WhatsApp Chatbot (OCR)** — Integrated OpenAI + Google Vision for prescription OCR via WhatsApp Business API. Production LLM integration with safety constraints.

## EDUCATION

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**B.Sc. Computer Engineering** · [Makerere University, Kampala](#)

2018 – 2022

## SKILLS

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**Proficient:** TypeScript, Node.js, JavaScript, APIs, distributed systems, microservices, GCP (Firestore, Pub/Sub, Cloud Functions, GCS), Kafka, NATS, gRPC, Kubernetes, Docker, MongoDB, SQL, Python

**Intermediate:** Go, AWS (S3, EC2, EBS), Rust, React, React Native, Redis (familiar)

**Languages:** English (native), Kiswahili